Same Photograph or Same Scene? Camera-Pose Forensics for Tracing Archival Photographs into Wartime Propaganda Print

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Abstract

Wartime propaganda magazines reprinted archival photographs after cropping, retouching, and recaptioning them; the record of that editorial labor survives only as pairwise identities between archival exposures and printed reproductions. Recovering those identities at scale is a verification problem, not a retrieval one: the historian must decide whether a print derives from the *same exposure* as an archival photograph, or from a second photograph of the same scene taken moments apart—the error that turns provenance tools into misattribution machines. We present a four-stage pipeline: self-supervised cross-domain retrieval, dense-correspondence filtering, homography inlier-ratio gating, and a new forensic signal—the relative camera pose inferred by a feed-forward 3D foundation model (VGGT). Re-editions of one exposure behave as pictures of a picture and yield near-zero inferred pose displacement; same-scene negatives do not. With all thresholds frozen on a development shard, the pose gate raises held-out verification precision from 83.7% (at 100% recall) to 90.2% at 98.4% recall, conditional on the candidate set. We release NCR-Match, 2,636 expert-verified photograph–print pairs with exposure-identity versus same-scene labels drawn from the North China Railway Archive, a CC BY 4.0 occupation-era collection, and evaluate cross-system transfer to an opposing propaganda apparatus and historian-in-the-loop utility.

Introduction

In March 1942 the *Jinchaji Pictorial*, the flagship photographic organ of the Chinese Communist base areas, printed a photograph by Sha Fei of the peasant Liu Hanxing enlisting in the Eighth Route Army, under the page title "Model Families: Parents Send Their Children, Wives Send Their Boys." The same exposure was reprinted at least three more times over the next four years; by 1945 Liu's name had been removed and the caption read "common people," and by 1946 he had become simply "a youth" (Du, Le, and Honig 2024; Du 2025). No single page records this trajectory. The editorial labor of wartime propaganda—cropping, retouching, splitting, recaptioning—is invisible in the printed object and undocumented by the institutions that performed it: as

[TODO: Figure 1: one verified photo–halftone reprint pair vs. one same-scene-seconds-apart negative, with inferred camera frusta overlaid — render from the pose-validity visualization batch (W11, report section 6.5)]

Figure 1: A verified reprint pair (one exposure, re-edited into print) versus a same-scene negative exposed moments apart, with the inferred cameras overlaid.

Du observes, such archives preserve the photographs but not the decisions (Du 2025). Propaganda systems operate precisely by recirculating a stable stock of authorized images under shifting textual frames (Yurchak 2005), so the record of that work survives only as thousands of pairwise identities between an archival exposure and a printed reproduction—each individually minor, collectively the only evidence of how the apparatus curated reality, what Du calls "the evidence of the small" (Du 2026). The scale defeats close reading alone (Arnold and Tilton 2019): the North China Railway Archive, the photographic record of the Japanese-run railway company at the center of occupation publicity in North China (Kushner 2006; Earhart 2008), holds roughly 39,000 photographs (1939–1945), an unknown fraction of which resurfaced in the coeval propaganda pictorial *North China* (*Hokushi*, 1939–1943). Figure 1 shows the pair type at issue.

Computer vision appears to be the obvious remedy, and published work has shown that learned embeddings can surface photograph–print candidates across the severe domain gap of halftone screening, contrast loss, and cropping (Du, Le, and Honig 2024). But retrieval answers the wrong question. Retrieval finds look-alikes; the historian's claim—*this print is that negative*—is a claim of *exposure identity*. Press photographers worked in bursts, and archives are dense with frames taken seconds apart from nearly the same position; a system that cannot separate "same photograph, edited" from "same scene, next frame" does not merely lose precision points, it manufactures confident misattributions of photographer, place, and date. The failure mode is not hypothetical: the second round of our benchmark's verification protocol excluded a nontrivial fraction of first-round candidates as exactly such same-scene negatives (see Dataset Construction). The distinction itself is classical—image phylogeny separates near-duplicates (one exposure, a chain of transformations) from semantically similar images of the same scene (Dias, Goldenstein, and Rocha 2013; Chum et al. 2007)—yet modern copy-detection benchmarks are born-digital and treat same-scene shots as negatives by construction (Douze et al. 2021; Pizzi et al. 2022; Wang, Sun, and Yang 2023), and no public benchmark poses this boundary on archival photo-to-print material. Where Du, Le, and Honig (2024) demonstrated proof-of-concept circulation forensics with retrieval alone, this paper contributes the verification decision retrieval cannot supply, delivered at benchmark scale and released with the submission. Nor is the problem antiquarian: recaptioned reuse of authentic photographs is the dominant mechanism of contemporary out-of-context misinformation, for whose modern form this community has built detectors and benchmarks (Aneja, Bregler, and Nießner 2023; Qi et al. 2024). A wartime propaganda corpus is the same mechanism with ground truth—a closed, fully attributable natural experiment in institutional image reuse, eighty years before the feed.

We answer the exposure-identity question with a four-stage funnel (Figure 2). A self-supervised fusion head over frozen vision backbones (DINOv3 (Siméoni et al. 2025), EfficientNetV2, Swin) retrieves top-ranked print candidates for each archival photograph; ASpanFormer (Chen et al. 2022) extracts dense correspondences and discards pairs with too few; a seeded robust homography fit (MAGSAC++) gates on the inlier ratio and aligns the photograph onto the print; and a final judge inspects a signal new to provenance analysis: the relative camera pose inferred by the feed-forward 3D foundation model VGGT (Wang et al. 2025a). The intuition is geometric. A genuine reprint is a picture of a picture: after alignment the pair carries no parallax, and the model infers essentially the same camera for both frames. A second exposure of the same scene, however similar, retains parallax that surfaces as rotation and lateral translation of the inferred camera. We convert this into a component-weighted pose-displacement score—strict on rotation and x/y translation, lenient on the depth and field-of-view terms that cropping and rescanning perturb—and threshold it. While feed-forward 3D foundation models have been used to verify whether two images depict the same surface or place (Cai et al. 2023; Xiangli et al. 2025; Maggio and Carlone 2026), and copy detection has relied on 2D appearance descriptors with same-scene shots treated as negatives by construction (Douze et al. 2021; Pizzi et al. 2022; Wang, Sun, and Yang 2023), we are—to our knowledge—the first to use the relative camera pose inferred by such models as a provenance signal separating re-editions of the same exposure (crops, edits, halftone reprints) from distinct exposures of the same scene. On the held-out validation shard of our benchmark, with every threshold frozen on the development shard, the full system attains verification precision 90.2% and recall 98.4% (F_1 94.1), conditional on the candidate set; the pose gate contributes $+6.5$ points of precision over inlier-ratio verification alone (83.7% $\rightarrow$ 90.2%) at a cost of 1.6 points of recall.

This paper makes five contributions:

- **Task.** We formalize *exposure-identity vs. same-scene discrimination* for archival photo-to-print tracing, grounding it in the phylogeny dichotomy between near-duplicate and semantically similar images (Dias, Goldenstein, and Rocha 2013) and the IND/NIND (identical vs. non-identical near-duplicate) taxonomy (Chum et al. 2007; Jinda-Apiraksa, Vonikakis, and Winkler 2013), and argue that it is the decision on which archival provenance claims—and their modern out-of-context analogues—rest.
- **Method.** Inferred relative camera pose as a provenance signal, with the component-weighted decision rule above and ablations isolating it against verification without pose, classical pose recovered for free from the already-computed homography and from essential-matrix decomposition, MASt3R pose (Leroy, Cabon, and Revaud 2024), and a learned classifier over identical inputs **[TODO: remaining Table 3 rows: B8 (essential-matrix pose), B9 (MASt3R, run W6), B11–B14 (alignment and matcher perturbations, runs W7/W8), plus all Shard-3 columns]**.
- **Benchmark.** NCR-Match: 2,636 verified photograph–print pairs linking the North China Railway Archive (~39,000 photographs, 1939–1945) to the magazine *North China* (1939–1943), embedded in a working corpus of 42,773 cropped images, curated with a two-round verification protocol in which same-scene near-misses are excluded as labeled hard negatives. Archive images and metadata are CC BY 4.0, so the benchmark is deposited with this submission with attribution to the archive. To our knowledge it is the first public benchmark with exposure-identity versus same-scene labels on cross-domain (photograph vs. halftone print) archival material. Labels are expert-adjudicated under a written codebook **[TODO: inter-annotator agreement: Cohen's κ from the dual-annotation campaign, report section 6.6]**.
- **Cross-system transfer.** With all thresholds frozen, the pipeline transfers from Japanese occupation propaganda to the opposing Communist base-area system: the *Jinchaji Pictorial* / Papers of Sha Fei corpus (2,146 photographs, 1,122 pages, 4,390 crops, 230 annotated pairs) studied by prior retrieval-only work (Du, Le, and Honig 2024). **[TODO: Retrieval Recall@10/15 and mAP over the 230 verified pairs + verification P/R/F1 conditional on hand-labeled filter survivors, thresholds frozen — Jinchaji run W10. Jinchaji side releases annotations and identifiers only, no images.]**
- **Historian-in-the-loop evaluation and findings.** An annotation reliability study **[TODO: κ, adjudicated gold labels, 'Unsure' as a first-class outcome]**, a measured workflow-impact analysis (review-queue reduction and expert adjudication throughput **[TODO: adjudication timing from the Shard-3 campaign]**), and new historical findings surfaced by the system **[TODO: 2–3 case**

studies from the confirmed-match list, adjudicated by the domain expert].

The system is built for a deployment path rather than a demonstration: the anonymized code in the supplementary material, the benchmark, and the annotation codebook are deposited with this submission, and the pipeline is designed to slot into the verification workflows of the archives and research groups that steward these collections. Concretely, the candidate-review interface used in the historian study of the Experiments section is part of the code release; the verified-pair layer is exported in a tabular, identifier-keyed format suitable for ingestion by an archive's public catalog; and the frozen-threshold transfer protocol demonstrated on the Jinchaji corpus doubles as the documented onboarding recipe for applying the system to a new collection. Its design is ethics-aware by principle: *the pipeline does not interpret; it retrieves* (Du 2026). Every accepted pair ships with its geometric evidence, every threshold is published, "Unsure" is reported rather than silently dropped, and the interpretive claim—what a recaptioned reprint *meant*—remains the historian's to make. We return to the dual-use risks of automated provenance assertion, and to the contextualization owed to imagery produced by occupation and propaganda apparatuses, in the Ethics section.

Related Work

Content-Based Retrieval for Archives

Deep metric learning made instance-level retrieval practical (Schroff, Kalenichenko, and Philbin 2015), and self-supervised contrastive objectives such as SimCLR proved effective for learning transformation-invariant descriptors (Chen et al. 2020). Transformer encoders sharpened global retrieval features further (Dosovitskiy et al. 2021; El-Nouby et al. 2021); today, self-supervised and image–text foundation encoders—DINOv2 and DINOv3 (Oquab et al. 2024; Siméoni et al. 2025), CLIP (Radford et al. 2021), and SigLIP 2 (Tschannen et al. 2025)—supply strong retrieval features off the shelf. Matching across a change of medium or style is the harder, cross-domain variant of the problem (Shrivastava et al. 2011; Hu, Zhang, and Lee 2023), and the photograph-to-halftone gap studied here is an extreme instance of it. We therefore treat retrieval as a component rather than a contribution: our fused encoder exists to propose candidates, and we evaluate it against the off-the-shelf encoders above.

Copy Detection and the Same-Scene Boundary

Near-duplicate and copy detection is the retrieval special case closest to our task. Early benchmarks were small and gentle: Copydays covers 157 source images under simple edits (Douze et al. 2009). DISC21 scaled the problem to a million-image reference set under adversarial transformations, but its queries and references remain born-digital (Douze et al. 2021), and SSCD distilled the resulting lessons into the standard copy-detection descriptor (Pizzi et al. 2022). The frontier these efforts name is precisely our problem class: the NDEC benchmark introduced hard negative pairs that are inherently similar rather than edited copies, with an asymmetrical similarity objective to handle them (Wang, Sun, and Yang 2023), and AnyPattern extends copy detection to unseen edit patterns (Wang et al. 2026). The distinction itself is older. Near-identical image detection was posed by Chum et al. (2007), and California-ND annotates a personal photo collection with exactly the split we need: identical near-duplicates (variants of one exposure) versus non-identical near-duplicates (different exposures of similar content) (Jinda-Apiraksa, Vonikakis, and Winkler 2013). Most directly, the image-phylogeny literature formalized the boundary over a decade ago: Dias, Rocha, and Goldenstein reconstruct transformation trees over near-duplicates (Dias, Rocha, and Goldenstein 2012) and, in extending trees to forests, define the *semantically similar* images that must be kept out of a phylogeny—the same scene captured at a "different point in space and time" (Dias, Goldenstein, and Rocha 2013). Provenance analysis operationalizes phylogeny at scale with 2D local features and appearance descriptors (Moreira et al. 2018), standardized by the NIST Media Forensics Challenge protocols (Guan et al. 2019) and recently rebuilt on vision transformers (Zhang et al. 2024; Lai et al. 2025). Across this entire line, the copy-versus-scene boundary is enforced by appearance alone; no system consults the camera geometry that defines it. To our knowledge, ours is the first provenance system to enforce the phylogeny boundary geometrically rather than by appearance, and the first tested on cross-domain, photograph-to-print archival material.

Geometric Verification and 3D Foundation Models

Confirming that a retrieved candidate corresponds to a query, rather than merely resembling it, is classically the job of local features and robust estimation: SIFT correspondences filtered by RANSAC (Lowe 2004; Fischler and Bolles 1981). Detector-free transformer matchers such as LoFTR (Sun et al. 2021) and ASpanFormer (Chen et al. 2022) find dense correspondences under weak texture and heavy degradation, where halftone dot screens defeat classical detectors; LightGlue (Lindenberger, Sarlin, and Pollefeys 2023) and RoMa (Edstedt et al. 2024, 2025) push accuracy and efficiency further. In our pipeline this stage is a learned form of the historian's side-by-side comparison. A newer family of feed-forward 3D foundation models regresses scene geometry directly: DUSt3R and MASt3R predict pointmaps and matches from image pairs (Wang et al. 2024; Leroy, Cabon, and Revaud 2024), Reloc3r regresses relative camera pose at scale (Dong et al. 2025), and VGGT jointly predicts cameras, depth, and point tracks for arbitrary image sets (Wang et al. 2025a). Deciding per image pair whether a homography (no parallax) or a fundamental matrix (genuine two-view parallax) better explains the correspondences is a classical model-selection problem (Torr 1998); our pose gate is its forensic descendant, replacing information criteria over sparse correspondences with halftone-robust dense matching and the metric pose head of a 3D foundation model. The nearest neighbors of that use are Doppelgangers and Doppelgangers++, which learn to reject visually convincing matches between *different* surfaces to disambiguate structure from motion (Cai et al. 2023; Xiangli et al. 2025); our decision has the opposite polarity—both images genuinely show the same surface, and the question is whether they are the same exposure of it. VGGT-SLAM 2.0 uses VGGT internals as a training-free verifier of candidate loop closures (Maggio and Carlone 2026): same-place verification inside SLAM, not duplicate-versus-reshot forensics. Computational re-photography estimated the pose delta between a historical photograph and a live viewfinder to guide a photographer back to the original viewpoint (Bae, Agarwala, and Durand 2010)— inter-photograph camera pose as a first-class quantity, but in an interactive rather than forensic setting. Same-photo discrimination itself appears once before, on 2D appearance: detecting whether two face crops derive from one photograph, to expose bias in kinship benchmarks (Dawson, Zisserman, and Nellåker 2018)— no geometry, and never extended to provenance. A related intuition, that a photograph of a flat print or screen betrays itself geometrically, underlies recapture detection in image-authenticity verification (Vilesov et al. 2024); that family makes single-image attack decisions, not pairwise provenance decisions. In sum, none of these lines uses inferred relative camera pose to decide exposure identity; that use, argued in the Introduction, is ours.

Why Not Sensor Forensics?

The natural forensic reflex for the same-exposure question is sensor-noise attribution: PRNU fingerprints identify the source camera of a digital image (Lukáš, Fridrich, and Goljan 2006) and survive, weakly, into printed copies (Goljan, Fridrich, and Lukáš 2008). Three facts rule this channel out here. First, wartime photographs were exposed on film; there is no digital sensor and hence no PRNU to extract. Second, a device fingerprint identifies the camera, not the exposure: two frames taken seconds apart with the same camera share it, so it answers the wrong question. Third, halftone screening and generational duplication destroy the high-frequency signal the method depends on, a degradation documented in the printed-image study itself (Goljan, Fridrich, and Lukáš 2008). Geometry survives exactly where sensor noise dies: the projective relation between two prints of one exposure is untouched by screening.

Historical Image Reuse and Cross-Domain Datasets

Reuse of visual material predates photography: the cost of woodblocks and copperplates led early modern printers to recycle blocks across editions, and object detection plus visual search now recovers those block-sharing networks (Lang 2025; Dutta, Bergel, and Zisserman 2021); contrastive training likewise clusters recurring printing ornaments across eighteenth-century books (Wang et al. 2025b). For photographic archives, Newspaper Navigator established the page-level visual-extraction lineage from which our cropping stage descends (Lee et al. 2020), and transductive fine-tuning has been applied to near-duplicate detection in scanned photo collections (Net et al. 2023)—the archival near-duplicate setting, addressed with 2D descriptors and no geometry. Purpose-built historical corpora are growing: EyCon offers roughly 130k digitized photographs and press pages from the conflicts of 1890–1918 (Giardinetti et al. 2024); the Illustrated London News dataset pairs some 72k wood engravings with multimodal search (Smits et al. 2025); and FORBIN releases a press-photography agency archive of the early twentieth century with rich archival metadata and an explicit image-reuse research agenda (Chelali et al. 2026). The closest methodological neighbor is the EyCon group's halftone similarity work (Aissi et al. 2025), which combines visual and textual similarity to improve similarity search over halftone press photographs. Two differences position our task against theirs: their goal is intra-corpus similarity neighborhoods within a single digitized collection, whereas ours is open-set matching across two collections—an archive of original photographs and a printed magazine; and their output is a ranked neighborhood, whereas ours is a verified pair-level decision about exposure identity. None of these corpora carries verified pair ground truth linking a photograph to its printed reproduction; NCR-Match fills that gap (see the contributions above). Finally, Du, Le, and Honig (2024) demonstrated proof-of-concept circulation forensics on WWII East Asian pictorials with an ensemble retrieval model, reporting 77.8% top-15 retrieval on the Jinchaji corpus, and Du traced the editorial cropping and recaptioning of such photographs through close reading (Du 2025). Building on that proof of concept, this paper contributes what neither work possessed: a scaled, benchmarked, and released matching system, and a verification signal that decides whether two images share an exposure.

Out-of-Context Misuse and Computational Propaganda

The decision our system makes—same exposure or not—is the evidentiary substrate of a problem this community already studies. Out-of-context misuse, in which an authentic image recirculates under a false caption, is the cheapest and most common form of visual misinformation; COSMOS detects it by grounding captions against image regions (Aneja, Bregler, and Nießner 2023), and SNIFFER adds multimodal LLM-based explanation (Qi et al. 2024). Media scholarship documents the same mechanics across a century: iconic photographs are revived and resignified as memes (Boudana, Frosh, and Cohen 2017), and their online circulations can be traced at the scale of hundreds of thousands of reposts (Smits and Ros 2023). A 1943 propaganda magazine recaptioning an archival photograph is mechanically the same act as a modern out-of-context post—the wartime corpus is a fully labeled natural experiment in institutional image reuse, with the added difficulty that each "repost" passed through a halftone press. Methodologically, we invert distant viewing (Arnold and Tilton 2019): rather than reading a large corpus for aggregate patterns, Digital Historical Forensics uses scale to locate the singular object and then reads it closely (Du 2026), with the division of labor between pipeline and historian fixed by the design principle stated in the Introduction.

Task and the NCR-Match Benchmark

Exposure Identity vs. Same Scene

Given a query archival photograph q and a candidate reproduction c cropped from a printed page, the task is to decide whether the pair is *exposure-identical*: whether both images derive from a single release of the shutter—one negative—so that c is related to q by a chain of reproduction transformations (cropping, rotation, mirroring, rescaling, retouching, halftone screening, and generational re-photography of prints). The contrasting case is a *same-scene negative*: q and c depict the same scene but originate from distinct exposures—adjacent frames on a roll, or frames made moments apart from a nearby vantage point. The dichotomy itself is established. Image phylogeny distinguishes near-duplicates, related by a transformation chain from a common source, from "semantically similar" images captured by a camera at a different point in space and time (Dias, Rocha, and Goldenstein 2012; Dias, Goldenstein, and Rocha 2013), and near-duplicate retrieval separates identical from non-identical near-duplicates (IND vs. NIND) (Chum et al. 2007; Jinda-Apiraksa, Vonikakis, and Winkler 2013). NCR-Match operationalizes this boundary as a labeling standard on cross-domain archival material, where it carries evidentiary weight: for the provenance claim "this archival photograph is the one printed on this page," a same-scene negative is not a weaker match but a false one.

Archival practice makes such negatives dense rather than exotic. Photographers on assignment exposed sequences of near-identical frames, and picture editors selected among the variants, sometimes printing different frames of one sequence in different issues (Du 2025); wartime pictorials also recirculated the same negative across publications (Du, Le, and Honig 2024). The confusable pairs in this corpus therefore arise organically from production practice, unlike born-digital copy-detection benchmarks, whose negatives are unrelated by construction and whose same-scene confusables are rare or synthetic (Douze et al. 2021; Wang, Sun, and Yang 2023). On our development shard, 316 of 641 expert-labeled candidate pairs (49%) are negatives, in large part same-scene look-alikes (**[TODO: exact same-scene vs. unrelated split of labeled negatives, from the negative-type audit of Shards 1–3]**). Table 1 summarizes the benchmark and its shards.

Source material	
Archive photographs (NCR Archive, 1939–1945)	∼39,000
North China magazine issues (1939–1943)	**[TODO: issue count]**
Working corpus (YOLOv11 crops, both collections)	42,773
Exposure-identity verification (two rounds)	
First-round candidate pairs	**[TODO: count]**
Excluded in round two as same-scene	**[TODO: count]** (∼7%)
Verified pairs (NCR-Match)	2,636
Labeled candidate pairs (pos. / neg. / Unsure)	
Shard 1 (development)	325 / 316 / 2
Shard 2 (validation, thresholds frozen)	252 / 386 / 0
Shard 3 (test, evaluated once)	**[TODO: n / n / n]**
Inter-annotator agreement (Cohen's κ)	**[TODO: κ]**
Transfer domain (Jinchaji Pictorial / Papers of Sha Fei)	
Photographs / magazine pages	2,146 / 1,122
Crops / annotated pairs	4,390 / 230

Table 1: NCR-Match at a glance. Verification labels exist for candidate pairs that survive retrieval and the correspondence floor; verification metrics are therefore conditional on the candidate set (see Experiments), and end-to-end recall is reported separately as a survival analysis over the 2,636 verified pairs.

Dataset Construction

Sources. NCR-Match pairs photographs from the North China Railway Archive—approximately 39,000 photographs taken in Japanese-occupied North China between 1939 and 1945, digitized and published with metadata under CC BY 4.0 (Center for Open Data in the Humanities (CODH), ROIS 2019)[1]—with reproductions in the Japanese-language pictorial *North China* (*Hokushi*, 1939–1943). Magazine pages mix several photographs with text; following the page-segmentation lineage of Newspaper Navigator (Lee et al. 2020), a YOLOv11 detector (Khanam and Hussain 2024) trained on a small annotated page sample crops individual photographs. The working corpus comprises 42,773 cropped images across the two collections; crop-stage quality is **[TODO: detection precision/recall on a 100-page manually audited sample — YOLOv11 QC audit]**.

Two-round verification. Candidate photo–print pairs were proposed by embedding retrieval with geometric matching over the working corpus (**[TODO: first-round candidate-pair count]**) and verified by hand in two rounds: round one screened candidates for visual identity; round two re-examined every accepted pair specifically for the same-scene failure mode. About 7% of first-round candidates were identified in round two as same-scene negatives and excluded, leaving 2,636 verified exposure-identity pairs, each carrying magazine issue, date, and page metadata. The exclusion rate is itself evidence for the task: even under careful single-pass review, roughly one accepted candidate in fourteen was a different exposure of the same scene. The excluded pairs are retained as labeled same-scene hard negatives (**[TODO: confirm the round-two exclusion list ships in the release]**).

Labeled verification shards. For threshold development and testing, the working corpus is deterministically partitioned into six shards of roughly 7,000 images each; three are labeled in this work. Within a shard, each query photograph's ten highest-ranked print candidates are matched with ASpanFormer (Chen et al. 2022), and pairs passing the correspondence floor (roughly 650 per shard; see Method) receive human labels. Verification labels thus exist only for pairs the funnel surfaces, and all verification metrics are reported conditional on the candidate set (see Experiments). Shard 1 is the development set—all thresholds and the pose-component weighting were chosen here—with 325 positive and 316 negative pairs. Shard 2 is held-out validation, evaluated once with everything frozen: 252 / 386. Shard 3 is an untouched test shard, labeled under the codebook below and evaluated exactly once: **[TODO: Shard-3 positive/negative/Unsure counts, from the Shard-3 labeling campaign]**. Unsure labels (**[TODO: Unsure counts per shard, from the dual-annotation campaign]**) are reported, never silently dropped.

[1] https://codh.rois.ac.jp/north-china-railway/

Transfer domain. A second, disjoint corpus supports cross-domain evaluation: the *Jinchaji Pictorial* and the Papers of Sha Fei (2,146 photographs, 1,122 magazine pages, 4,390 crops), with 230 annotated photo–print pairs from Du, Le, and Honig (2024). The two corpora come from opposing propaganda systems—Japanese occupation publishing and Communist base-area publishing—with different printing technology and degradation profiles.

Annotation Protocol and Reliability

Labels are governed by a written codebook whose rulings fix the two genuinely contested cases: adjacent frames from one photographing sequence are same-scene negatives, however similar; and disjoint crops from a single negative are positives—exposure identity does not require overlapping image content. Each candidate pair is labeled independently by two annotators, a domain-expert historian of wartime East Asian print media and a trained annotator, with *Unsure* as a first-class label; disagreements are adjudicated into gold labels. Inter-annotator agreement is **[TODO: Cohen's κ over the dual, independently labeled Shard 1–3 candidate pairs ($\approx$1,950), from the relabeling campaign]**. Every verified positive additionally carries a trace type—halftone-only, crop, rotation/mirror, retouch, or reception—with distribution **[TODO: trace-type distribution over the 2,636 verified positives, expert taxonomy tagging]**; per-type pipeline pass rates appear in the Experiments section. The recaption class— near-identical pixels under a changed caption—is detectable as an image match, but its editorial significance is invisible to an image-only system (see Limitations).

Release, Licensing, and Known Biases

NCR-Match is deposited with the supplementary material, together with anonymized code, rather than promised: image crops, pair labels, shard assignments, trace annotations, the codebook, and per-pair metadata. The North China Railway Archive publishes its photographs and metadata under CC BY 4.0, and the release attributes all NCR imagery to the archive accordingly. For the Jinchaji domain we release annotations and image identifiers only, no image data. A datasheet (Gebru et al. 2021) documents collection, the two-round verification protocol, annotation, and known biases. Chief among these: the source archive is the product of Japanese corporate and military photography—it records what an occupation apparatus chose to photograph—so both subject matter and the trace-type distribution reflect that selection. We state this as a documented limitation on generalization claims rather than treating the corpus as a neutral sample of wartime photography (see Ethics).

Method

Figure 2 gives an overview. Given an archival photograph (source) and a corpus of cropped print images (targets), the pipeline is a four-stage funnel: (1) *fusion retrieval* proposes, for each source, its top-K most similar targets; (2) *detector-free matching* establishes dense correspondences and discards pairs below a correspondence floor; (3) a *geometric-consistency gate* requires the correspondences to be explainable by a single planar projective map; (4) a *pose-based decision layer* labels each surviving pair *same exposure* or *same scene, different exposure*. Stages are ordered by cost, and each stage writes every measured signal, decision, and machine-readable rejection reason to an append-only manifest, so all threshold-level ablations replay from cached artifacts without GPU recomputation; every randomized estimator is seeded, and the released code reproduces the reported decisions bit-for-bit. Because the funnel restricts what later stages see, every metric downstream is reported against a named denominator (see Experiments).

[TODO: pipeline figure: four-stage funnel with one photo–print pair as running example (archival photograph, retrieved print crop, ASpanFormer correspondences, homography-aligned overlay, VGGT pose deltas), annotated with per-stage survivor counts from the survival audit (W5/W11)]

Figure 2: Overview of the four-stage verification funnel.

Stage 1: Fusion Retrieval

The retrieval encoder follows the frozen-ensemble design validated on archival photo–print retrieval by Du, Le, and Honig (2024), with the ViT backbone upgraded to DINOv3. Each image is resized to 224×224 and rendered as three-channel grayscale before ImageNet normalization: across the photograph/halftone gap, tonal structure, not color, is the informative channel. Three frozen backbones encode the image—DINOv3 ViT-L/16 (1,024-D; Siméoni et al. 2025), EfficientNetV2-M (1,000-D; Tan and Le 2021), and Swin-T (1,000-D; Liu et al. 2021)—and their concatenated 3,024-D output is projected by a two-layer MLP head ($3{,}024 \to 2{,}000$, ReLU, $\to 1{,}000$). Only the head is trained, with SimCLR-style self-supervision (Chen et al. 2020) on the unlabeled 42,773-crop working corpus: two augmented views (random crops and rotations) per image, contrastive temperature 0.05, batches of 16 images (32 views), 50 epochs, SGD with learning rate 10^{-5}, momentum 0.9, weight decay 10^{-3}, fixed seed. No pair labels are used at any stage of training. Because the corpus mixes both domains, the head learns halftone-invariant features; because it is also the corpus later searched, the setup is transductive, which we disclose and control **[TODO: transductive control: retrain the fusion head excluding all Shard-3 images and show Shard-3 verification F1 unchanged within the bootstrap CI — run W9]**. Embeddings are ℓ_2-normalized and compared by cosine similarity; the $K{=}10$ highest-scoring targets per source become candidate pairs **[TODO: retrieval Recall@K (K=5–30) and mAP over verified pairs, per encoder incl. off-the-shelf baselines, Table 2 — run W4 from the cached `feature_cache.npz`]**.

Stage 2: Detector-Free Matching and Correspondence Floor

Halftone screening, retouching, and paper degradation defeat detector-based local features, so we match with ASpanFormer (Chen et al. 2022) (pretrained outdoor weights; grayscale; long side resized to 1,024 px, aspect preserved), which regresses dense correspondences without keypoint detection. For each candidate pair we record the raw correspondence set and its epipolar-consistent subset under a fundamental-matrix fit with seeded RANSAC (`FM_RANSAC`, 1 px reprojection threshold, `cv2.setRNGSeed` before every robust fit; Fischler and Bolles 1981; Bradski 2000). A pair advances only if its correspondence count clears a floor τ_{corr} **[TODO: BLOCKING, resolve before any other table row is run: state τ_{corr} for the Shard-1/2 runs (working log records 50/75/100 in different places), resolve whether the floor applies to raw correspondences or to epipolar inliers (the code gates on the filtered count, the log narrative says raw), and verify that the resolved definition regenerates exactly the 641/638 labeled candidate sets of Shards 1–2 that produced rows B2/B3; row B1 sweeps the floor over the full observed correspondence range on the development shard (best-F1 floor 2,375 raw correspondences, applied frozen to Shard 2), CPU-only from `aspan_all_manifest.jsonl`]**. The floor is a compute gate, not a match decision: it keeps the expensive pose stage off hopeless pairs, and because every attempted pair—including failures—is logged, the floor can be swept post hoc without re-running the matcher.

Stage 3: Geometric-Consistency Gate

From the stored raw correspondences we estimate a source→target homography H with seeded USAC_MAGSAC (5 px threshold; Barath et al. 2020) and define the inlier ratio ρ as the fraction of raw correspondences consistent with H. If two images are re-editions of one exposure, the mapping between them is, to good approximation, a single planar projective transform—cropping, rescanning, rotation, and page geometry compose into one homography—so ρ is high. Correspondences that scatter off a single homography indicate either scene-level similarity with parallax or unstable matching. The gate accepts pairs with $\rho \geq 0.65$, derived on the development shard and frozen thereafter. A deliberately lenient sanity pre-pass (finite, non-singular H; minimal correspondence, inlier, and warped-overlap requirements) skips the pose stage only for egregiously degenerate alignments, which are recorded as rejections.

Stage 4: Pose-Based Exposure-Identity Decision

The homography then serves a second purpose: the source is warped onto the target canvas, and the aligned pair is passed jointly to VGGT (Wang et al. 2025a) **[TODO: pin the exact VGGT checkpoint, commit hash, and inference resolution in the reproducibility checklist]**, a feed-forward 3D foundation model that regresses for each frame a nine-dimensional camera encoding $[\mathbf{t}, \mathbf{q}, \mathbf{f}]$: translation $\mathbf{t} \in \mathbb{R}^3$ in the model's normalized scene units, an orientation quaternion $\mathbf{q}$, and vertical/horizontal field of view $\mathbf{f} \in \mathbb{R}^2$. From the two frames' encodings we take four deltas: the rotation angle $\delta_{\text{rot}} = 2 \arccos|\langle \hat{\mathbf{q}}_s, \hat{\mathbf{q}}_t \rangle|$ (unit-normalized quaternions; the absolute dot product resolves the $\mathbf{q} \equiv -\mathbf{q}$ ambiguity), the lateral shift $\delta_{xy} = \|\mathbf{t}_s^{(xy)} - \mathbf{t}_t^{(xy)}\|_2$, the depth shift $\delta_z = |t_s^{(z)} - t_t^{(z)}|$, and the zoom shift $\delta_{\text{fov}} = \|\mathbf{f}_s - \mathbf{f}_t\|_2$. Each is scaled, capped, and weighted into a single score

$$S = \sum_{c \in \{\text{rot}, xy, z, \text{fov}\}} w_c \min\left(\frac{\delta_c}{s_c}, \gamma\right), \tag{1}$$

with weights $(w_{\text{rot}}, w_{xy}, w_z, w_{\text{fov}}) = (1, 1, 0.25, 0.25)$, scales $(s_{\text{rot}}, s_{xy}, s_z, s_{\text{fov}}) = (2°, 0.020, 0.100, 0.100)$, and per-term cap $\gamma = 3$. Here δ_{rot} is converted to degrees before scaling, and the translation deltas are measured in VGGT's normalized scene units, so each ratio δ_c / s_c is dimensionless. The asymmetry is the design: re-orientation and lateral viewpoint change are physical evidence that the camera moved—a different exposure—whereas apparent zoom and depth shifts routinely arise from cropping, enlargement, and reproduction even in true copies, so those terms contribute weakly and saturate **[TODO: empirical justification: per-component violin/ROC plots of $\delta_{\text{rot}}, \delta_{xy}, \delta_z, \delta_{\text{fov}}$ on exact-copy vs. same-scene labeled pairs (CPU, from `vggt_judged_manifest.jsonl`), Figure 3; component ablations rotation+xy-only vs. fov+z-only, Table 3 rows B4/B5]**. A pair is declared the same exposure iff

$$\rho \geq 0.65 \quad \wedge \quad S \leq 2.13, \tag{2}$$

both thresholds derived once on the development shard, frozen for all other data, and committed as named constants in the released code. VGGT's global register-token cosine similarity is recorded for every pair but is deliberately *excluded* from the rule: its distributions on exact copies and same-scene negatives overlap heavily and a gate on it adds no separation—a negative finding we report as such **[TODO: Table 3: global-similarity-gate baseline row B6 (cosine ≥ 0.90) plus the distribution-overlap figure, CPU from cached manifests]**.

Why Pose Can Judge a Picture of a Picture

An objection must be met head-on: an aligned exact copy is a photograph of a photograph—planar, degenerate two-view geometry, far from the multi-view scenes a 3D foundation model is trained on. Can its pose estimates be trusted there? Our rule requires far less than trustworthy metric pose. After Stage-3 alignment, a true re-edition presents the model with two frames related by approximately the identity: every change that reproduction introduces is photometric or planar-projective, there is no parallax left to explain, and the pose head regresses near-coincident cameras—near-zero δ_{rot} and δ_{xy}. A same-scene negative—a second exposure taken steps away or moments later—cannot be flattened this way: no single homography accounts for a genuine viewpoint change of a non-planar scene, residual parallax survives alignment, and the pose head absorbs it as relative camera motion. The signal is the degeneracy itself: we use the pose head as a parallax detector, not a camera localizer. Crucially, the near-zero pose response on true copies is a measured behavior of the system, not an assumption **[TODO: synthetic battery: ∼200 verified exact-copy pairs under controlled crops, rotations, halftone simulation, and mild perspective warps; report the fraction of pose scores remaining ≤ 2.13 per transform type — run W11] [TODO: measured pose-score distributions per class on the labeled shards, referenced from the Experiments section]**.

Design Alternatives

The pose stage is one of several ways to extract viewpoint evidence from signals the pipeline already computes, and we evaluate the alternatives under an identical protocol on the same candidate sets. Decomposing the Stage-3 homography yields a relative rotation and translation at no additional cost (Malis and Vargas 2007); the same correspondences support an essential-matrix pose under assumed intrinsics (Nistér 2004); MASt3R (Leroy, Cabon, and Revaud 2024) provides an alternative learned pose head; and a small learned classifier over the recorded per-pair signals (correspondence counts, ρ, the four pose deltas, global similarity), in the spirit of Doppelgangers-style match verification (Cai et al. 2023), replaces the hand-set rule with a fitted one. All four appear in Table 3 **[TODO: Table 3 rows B7 (homography-decomposition pose), B8 (essential-matrix pose), B9 (MASt3R pose, run W6), B10 (learned classifier); B7/B8/B10 are CPU-only from cached manifests]** alongside the component and threshold ablations.

Experiments

Our evaluation answers six questions, each with its own denominator: (1) does the learned retrieval front-end outperform off-the-shelf copy-detection and self-supervised descriptors on photo-to-halftone retrieval (Table 2); (2) does the inferred-pose gate improve exposure-identity vs. same-scene verification over cheaper geometric alternatives (Table 3); (3) do the two hand-set thresholds sit on plateaus rather than spikes (Figure 4); (4) what fraction of known reproductions the full funnel recovers end to end (Table 4); (5) does the frozen pipeline transfer to a second propaganda system; and (6) does the system measurably change historians' work.

Protocol

Split roles. The working corpus is partitioned into six shards by deterministic batching; three are labeled. **Shard 1 is the development set**: the inlier-ratio cutoff (0.65), the pose-score threshold (2.13), and the pose-component weighting itself were all derived here, and Shard-1 numbers should be read as development-set results. **Shard 2 is held-out validation**: it was evaluated once, with every parameter frozen, and never used for tuning. **Shard 3 is an untouched test set**, labeled under the written codebook of the Human Evaluation subsection and evaluated exactly once, with everything frozen **[TODO: single-shot Shard-3 evaluation (jobs W2/W3), scheduled after all thresholds and tables are locked]**. Labeled pairs: Shard 1, 325 positive / 316 negative (641); Shard 2, 252 positive / 386 negative (638); Shard 3 **[TODO: label counts after codebook annotation]**. Pairs labeled *Unsure* are excluded from these counts and reported as a first-class outcome, never silently dropped.

Denominator discipline. Every metric below names its denominator. *Retrieval* Recall@K and mAP are computed over the 2,636 verified photo–print pairs, with the full 42,773-crop working corpus as distractors: they answer whether a true reproduction, when one exists, surfaces in the top-K list. *Verification* precision, recall, and F1 are *conditional on the candidate set*: ground-truth labels exist for the candidate pairs that survive retrieval and the correspondence floor, and all metrics in Table 3 are computed over these expert-adjudicated pairs. We never quote a verification number as end-to-end performance. The paper's *only* end-to-end claim is the survival analysis of Table 4, whose denominator is again the 2,636 verified pairs.

Statistics, seeds, compute. Every P/R/F1 and PR-AUC cell carries a bootstrap 95% confidence interval (10,000 resamples over pairs) (Efron and Tibshirani 1993) **[TODO: add CIs to all table cells once rows complete]**. Variants are compared to the full system with exact McNemar tests on paired per-pair decisions (McNemar 1947), Holm-corrected within each table (Holm 1979). The final representation model is retrained with three seeds (mean±std) **[TODO: 3-seed retrain, job W12]**; ablation variants use a single disclosed seed. All RANSAC estimation is seeded (`cv2.setRNGSeed` before every call; fundamental-matrix RANSAC at a 1-pixel reprojection threshold) (Fischler and Bolles 1981), making the geometric stages bit-for-bit reproducible; a five-repeat seed check on Shard 1 confirms near-zero geometric-stage variance **[TODO: 5-seed repeat numbers]**. The fusion head is trained transductively (self-supervised, on the unlabeled evaluation corpus); a control retrain excluding all Shard-3 images is reported in **[TODO: transductive control, Shard-3 F1 within CI — run W9]**. Experiments ran on **[TODO: GPU model(s), VRAM, total GPU-hours per experiment family]**. The decision layer, with both thresholds as named constants, and scripts that regenerate every table from cached per-pair manifests, are part of the anonymized code in the supplementary material.

Retrieval: Candidate Proposal

Table 2 compares the retrieval front-end against frozen off-the-shelf descriptors under identical preprocessing, following the distractor-corpus protocol of the Image Similarity Challenge (Douze et al. 2021). A GeM descriptor (Radenović, Tolias, and Chum 2019) represents the classical landmark/instance-retrieval family (GeM/DELG) alongside the copy-detection and foundation-model encoders. Two rows isolate our design choices: DINOv3-alone with the SSL head (does the three-backbone fusion beat its best member?) and fusion without SSL (does the SimCLR-style head (Chen et al. 2020) matter?).

Verification: Exposure Identity vs. Same Scene

Table 3 is the paper's core table. All rows share the identical, fixed candidate sets described in the Protocol subsection, so differences are attributable to the decision rule alone; metrics

	Encoder	R@
A1	SSCD (Pizzi et al. 2022)	
A2	DINOv2 global (Oquab et al. 2024)	
A3	DINOv3 global (Siméoni et al. 2025)	
A4	CLIP / SigLIP 2 (Radford et al. 2021; Tschannen et al. 2025)	
A5	GeM (Radenović, Tolias, and Chum 2019)	
A6	ViT ensemble (Du, Le, and Honig 2024)	
A7	Ours: fusion + SSL	
A8	DINOv3-alone + SSL	
A9	Fusion, no SSL	

Table 2: Retrieval over the 2,636 verified pairs with the 42,773-crop corpus as distractors. **[TODO: All cells: W4 = featurize the corpus with each frozen encoder and score Recall@K/mAP; W9 = retrain rows A8–A9 (DINOv3-alone + SSL head; fusion without SSL). A6 uses the ensemble configuration of Du, Le, and Honig (2024), reimplemented from the published description (confirm weights are publicly released before claiming otherwise).]**

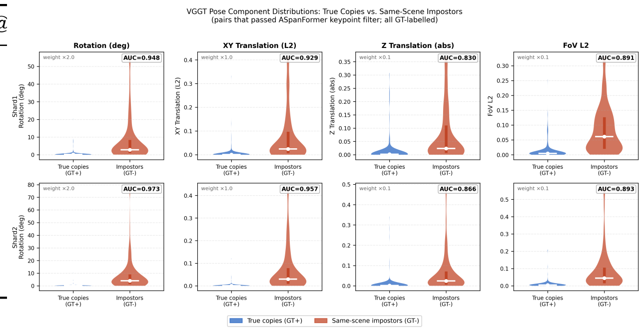

Figure 3: Per-component pose-delta distributions for exposure-identical pairs vs. same-scene negatives among correspondence-floor survivors. Exposure-identical pairs concentrate near zero on every component. Single-component AUCs — rotation 0.948 / 0.973, xy-translation 0.929 / 0.957, z 0.830 / 0.866, fov 0.891 / 0.893 (Shard 1 / 2) — support the weighting asymmetry of Eq. 1.

are conditional on the candidate set throughout. Row B0 is the accept-everything base rate, i.e. the class prior of the candidate set: precision 50.7 on Shard 1 and 39.5 on Shard 2 at recall 100. Row B2 (correspondence floor + inlier ratio, no pose) already reaches 82.9 / 97.2 / 89.5 (P / R / F1) on Shard 1 and 83.7 / 100 / 91.1 on Shard 2. Adding the component-weighted pose gate (row B3, the full system) removes 17 false positives at the cost of 3 true positives on Shard 1, and 22 false positives at the cost of 4 true positives on Shard 2, yielding 86.7 / 96.3 / 91.3 and 90.2 / 98.4 / 94.1. Because the pose gate only removes pairs from the inlier-gate survivors, the discordant-pair counts for B2 vs. B3 are fully determined, and the exact McNemar p-values are 2.6×10^{-3} (Shard 1) and 8.6×10^{-4} (Shard 2). After Holm correction across the five Shard-2 tests in Table 3, every reported difference except the rotation+xy row (B4, $p = 0.50$) remains significant at $\alpha = 0.05$.

Three groups of rows probe the mechanism. *Component rows* B4/B5 split the pose score of Eq. 1 into its high-weight terms (rotation angle and xy-translation) and its low-weight terms (fov shift and z-translation). The outcome matches the design: the low-weight gate rejects nothing at the published threshold — B5 is decision-identical to the pose-free rule B2 — while B4 alone is statistically indistinguishable from the full system ($p = 0.50$): rotation and lateral translation carry essentially all of the pose signal. The per-component distributions agree (Figure 3): single-component AUCs are 0.948 / 0.973 for rotation and 0.929 / 0.957 for xy-translation, against 0.830–0.893 for z and fov. Row B6 gates on the global feature-similarity signal that the pipeline records but *excludes* from its decision rule: a dev-tuned gate on it reaches only F1 77.5 / 72.6 with PR-AUC 0.853 / 0.838 — far below every geometric row — which is why it is recorded but not used. *Alternative-pose rows* B7–B9 answer the question a reader should ask: the alignment stage already estimates a homography, and decomposing it yields relative rotation and translation for free (Malis and Vargas 2007) (B7); an essential-matrix pose from the same correspondences is nearly as cheap (Nistér 2004) (B8); and MASt3R (Leroy, Cabon, and Revaud 2024) is the strongest alternative feed-forward pose head (B9). Each is tuned on Shard 1 only and applied frozen, exactly as our gate was. The classical alternative is genuinely competitive — the strongest form of the objection — but consistently weaker: at its dev-tuned $4.5°$ rotation threshold, B7 gives up 4.9 recall points on Shard 1 and 2.3 F1 points on Shard 2 relative to the full system, and its threshold-free PR-AUC trails the VGGT score by 4.2 and 3.1 points (0.922 / 0.962 vs. 0.964 / 0.993), a gap that is stable under $\pm 30\%$ focal-length misspecification. Row B10 replaces the hand-set thresholds with a logistic-regression classifier over the stored per-pair signals, in the spirit of Doppelgangers (Cai et al. 2023), trained on Shard 1 with a cross-validated threshold. The fitted rule trades recall for precision (98.0 / 95.6 vs. the full system's 90.2 / 98.4 on Shard 2), a significant paired difference ($p = 0.021$; Holm-adjusted 0.042). We nonetheless keep the two-threshold rule as the deployed default: it is interpretable, and it favors recall — in provenance work a missed pair is a discovery permanently lost, while a false positive is caught in adjudication — but B10 shows clear headroom for a learned decision layer. Rows B11–B14 perturb the geometry stage itself (alignment off; filtered vs. raw alignment correspondences; matcher swaps to LightGlue (Lindenberger, Sarlin, and Pollefeys 2023) and RoMa (Edstedt et al. 2024)).

Threshold Sensitivity

Both operating constants of Eq. 2 are hand-set, so we show they lie on plateaus. Figure 4 sweeps the inlier-ratio cutoff over $[0.30, 0.90]$ and plots F1 per shard with 0.65 marked (panel a); gives full ROC and PR curves over the continuous pose score with the 2.13 operating point marked (panel b); and shows a two-dimensional F1 heatmap over (inlier ratio, pose score) on the development shard with the chosen operating point, replotted on the untouched test shard to show that the plateau, not just the point, transfers (panel c). The frozen operating point was evaluated on Shard 3 exactly once, be-

	Variant	Shard 1 (dev) P / R / F1	Shard 2 (val) P / R / F1	Shard 3 (test) P / R / F1	PR-AUC (S2)	McNemar vs. B3 (S2)
B0	Accept-all (base rate)	50.7 / 100 / 67.3	39.5 / 100 / 56.6	**[TODO: S3]**	—	$< 10^{-6}$
B1	Correspondence floor only (dev floor 2,375)	88.4 / 89.2 / 88.8	87.8 / 91.7 / 89.7	**[TODO: S3]**	0.962	8.1×10^{-3}
B2	+ inlier-ratio gate (no pose)	82.9 / 97.2 / 89.5	83.7 / 100 / 91.1	**[TODO: S3]**	0.996	8.6×10^{-4}
B3	**+ weighted pose gate (full)**	**86.7 / 96.3 / 91.3**	**90.2 / 98.4 / 94.1**	**[TODO: S3]**	0.993	—
B4	Pose: rotation + xy terms only	86.2 / 96.3 / 91.0	89.5 / 98.4 / 93.8	**[TODO: S3]**	—	0.50
B5	Pose: fov + z terms only	82.9 / 97.2 / 89.5	83.7 / 100 / 91.1	**[TODO: S3]**	—	8.6×10^{-4}
B6	Global-similarity gate (dev thr. 0.955)	70.3 / 86.5 / 77.5	61.1 / 89.3 / 72.6	**[TODO: S3]**	0.838	—
B7	Homography-decomposition pose (dev 4.5°)	86.6 / 91.4 / 88.9	86.9 / 97.2 / 91.8	**[TODO: S3]**	0.962	**[TODO: frozen-thr. pairing]**
B8	Essential-matrix pose	**[TODO: five-point + RANSAC on cached ASpanFormer correspondences; CPU]**				
B9	MASt3R relative pose	**[TODO: W6: MASt3R pose on all labeled pairs, same tune-on-S1 protocol]**				
B10	Learned classifier (LR, train S1)	—	98.0 / 95.6 / 96.8	**[TODO: S3]**	0.993	0.021
B11	Alignment off	**[TODO: W8: VGGT on unaligned crops]**				
B12	Filtered alignment correspondences	**[TODO: W8: raw vs. filtered correspondences for alignment]**				
B13	Matcher swap: LightGlue	**[TODO: W7: replace ASpanFormer; re-tune both thresholds on S1, frozen on S2/S3]**				
B14	Matcher swap: RoMa	**[TODO: W7: same protocol as B13]**				

Table 3: Verification conditional on the candidate set: the expert-labeled filter-survivor pairs of each shard (641 on Shard 1, 638 on Shard 2, **[TODO: n]** on Shard 3). Thresholds and weights for every variant are tuned on Shard 1 only and applied frozen elsewhere. McNemar p-values are exact tests on paired decisions; Holm correction across the table **[TODO: apply once all rows are run]**. Shard-3 column filled by the single frozen evaluation **[TODO: W2/W3]**.

Stage	Pairs alive	% of 2,636
All verified pairs	2,636	100.0
Retrieval top-10	**[TODO: W5]**	**[TODO: W5]**
+ correspondence floor	**[TODO: W5]**	**[TODO: W5]**
+ inlier ratio ≥ 0.65	**[TODO: W5]**	**[TODO: W5]**
+ pose score ≤ 2.13 (end-to-end)	**[TODO: W5]**	**[TODO: W5]**

Table 4: Stage-by-stage survival of the 2,636 verified photo–print pairs at frozen thresholds **[TODO: W5: end-to-end recall audit; also report which stage dominates the loss]**.

fore any sweep; the Shard-3 heatmap is a post-hoc robustness analysis computed after that single frozen evaluation, with no feedback into any reported operating point.

End-to-End Survival of Verified Pairs

Verification metrics above say nothing about pairs the funnel never surfaces. Two survival analyses address this at different scopes. First, *within the labeled candidate sets*: of the 325 confirmed positives on Shard 1, the inlier gate removes 9 and the pose gate a further 3 (96.3% survive); of the 252 on Shard 2, the inlier gate removes none and the pose gate 4 (98.4%); jointly, 561 of 577 confirmed positives (97.2%) survive the two decision gates. These figures cannot speak to the retrieval and correspondence-floor stages, because labels exist only for pairs that already passed them. Second, therefore, the full-corpus audit: Table 4 pushes all 2,636 verified pairs through every stage at frozen thresholds and reports the fraction still alive after each. That table is the paper's only end-to-end claim, and its final row answers the historian's operative question: *of the reproductions known to exist, how many does the system find?*

Cross-Propaganda-System Transfer

To test whether the pipeline and its thresholds encode anything specific to one publication apparatus, we freeze everything — weights, thresholds, correspondence floor — and run the identical system on the Jinchaji Pictorial materials from the Papers of Sha Fei (2,146 photographs, 1,122 magazine pages, 4,390 crops, 230 annotated pairs): Communist base-area propaganda, produced under different printing technology and degradation than the Japanese-run *North China* corpus, and studied previously by Du, Le, and Honig (2024), whose system reached 77.8% top-15 retrieval on this collection. We report Recall@10/15 against that reference **[TODO: W10: frozen-pipeline Jinchaji run]** and verification P/R/F1, conditional on the candidate set, on hand-labeled filter survivors **[TODO: Jinchaji survivor labeling under the same codebook]**. For this collection the supplementary release contains annotations and image identifiers only; the images themselves are not redistributed.

Human Evaluation

Annotation reliability. NCR-Match itself was built under the two-round verification protocol described in Dataset Construction, whose second round excluded same-scene negatives — precisely the confusion the verification stage exists to resolve. For the shard labels used in Table 3, a written one-page codebook fixes the contested cases (adjacent contact-sheet frames; crops of a same-scene negative), after which two annotators — a domain-expert historian of the period and a trained research assistant — independently label all candidate pairs as positive, negative, or *Unsure*, without access to each other's answers. We report Cohen's κ (Cohen 1960) **[TODO: κ per shard from the dual relabeling]**, per-shard *Unsure* counts **[TODO: counts]**, and adjudicate disagreements into the gold labels used throughout.

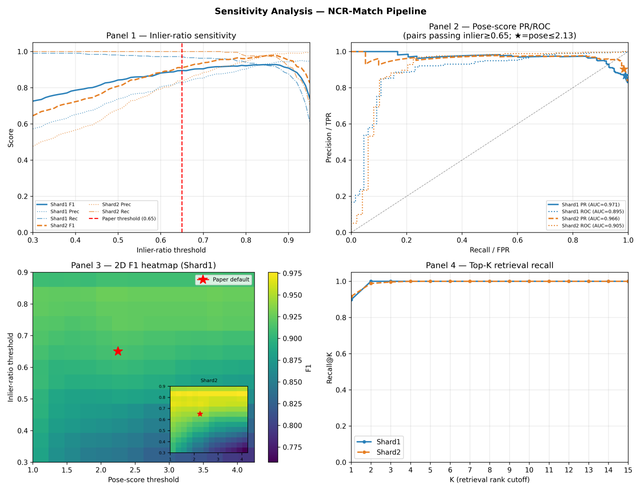

Figure 4: Threshold sensitivity of the two frozen operating constants. Top left: inlier-ratio sweep (0.65 marked). Top right: PR/ROC over the continuous pose score among inlier-gate survivors (PR-AUC 0.971 / 0.966; $\star$ = 2.13). Bottom left: 2-D F1 surface over both thresholds on the development shard ($\star$ = published operating point; Shard-2 inset) — a broad plateau; the published point deliberately sits on its recall-leaning side rather than at the F1 maximum. Bottom right: within-labeled-set Recall@K. **[TODO: replot the 2-D surface on Shard 3 after W2/W3]**

[TODO: hard positives accepted (heavy crops, coarse halftone, retouch) vs. hard same-scene negatives rejected, plus representative residual errors from the adjudicated Shard-2 error set (27 FP, 4 FN)]

Figure 5: Hard cases and residual failure modes.

Workflow impact. We quantify the system's contribution to the historian's workflow as reduction of the review queue, with named denominators. A manual baseline does not exist at this scale: exhaustively comparing $\sim$39,000 archive photographs against the 4,558 magazine crops is on the order of 1.8×10^8 pairwise judgements — at ten seconds per judgement, decades of person-time. The funnel reduces the roughly 70,000 top-10 candidate pairs of each labeled shard to 641 and 638 pairs requiring expert eyes, a two-orders-of-magnitude reduction, and the verification stage then concentrates truth within that queue: the class prior of the candidate set is 50.7% / 39.5% positive (Shards 1 / 2), whereas 86.7% / 90.2% of what the system finally asserts is a confirmed positive. Expert adjudication throughput on this queue is **[TODO: median seconds per pair, logged during the Shard-3 labeling campaign]**, so a shard's verified output costs hours of expert time rather than months. A controlled within-subjects user study with independent historians — tool vs. plain retrieval list, counterbalanced — is planned as future work; we do not claim measured time savings beyond the queue-reduction arithmetic above.

Qualitative Analysis and Failure Modes

Figure 5 shows hard positives the system accepts (heavy crops, coarse halftone screening, retouched reprints) alongside hard negatives it rejects (same scene, exposures seconds apart), and representative residual errors **[TODO: figure from adjudicated Shard-2 errors: 27 FP, 4 FN]**. We additionally tag every confirmed positive with a manipulation-trace type — halftone-only, crop, rotation/mirror, retouch, recaption-with-near-identical-pixels (Du 2026) — and report per-type pipeline pass rates **[TODO: taxonomy tagging of confirmed positives; per-type survival]**. One trace class is invisible to this system by construction: recaption-only reuse, in which the printed image is pixel-wise near-identical and only the surrounding caption changes. The pipeline correctly verifies such pairs as the same exposure — that is its function — but it cannot detect the caption substitution itself. Detecting textual and layout-level traces around a verified image pair is future work; the image-level verdicts reported here are the evidentiary substrate such a system would require.

Findings for Historians

The pipeline's product is not historical knowledge; it is a queue of adjudicated photo–print pairs, each carrying its geometric evidence (retrieval rank, inlier ratio, pose score, aligned overlay) and its provenance metadata (publication, issue, date, page). It uses scale to locate the singular object; the close reading of that object remains the historian's work (Du 2026). Findings of this kind have precedent at proof-of-concept scale: Du, Le, and Honig (2024) established the misattributed and mislocated Xifengkou/Futuyu photograph and the four-printing recaptioning chain of Liu Hanxing's enlistment portrait, and Du (2025) developed their historiographic implications. We report here what the scaled system adds: discoveries from the 42,773-crop working corpus that are, to our knowledge, previously undocumented in the scholarship on either archive, surfaced by the pipeline and adjudicated by a domain historian.

Finding 1 (a new version chain). **[TODO: Insert a NEW version chain from the adjudicated accepted-pair list: one North China Railway Archive photograph (give the archive identifier) reproduced in two or more magazine placements with materially different captions or layouts. Required elements: full provenance line for every printing (publication = North China (Hokushi), issue number, date, page); caption transcription and translation; per-pair pipeline evidence (retrieval rank, inlier ratio, pose score); one aligned-overlay figure. Must NOT be the Xifengkou/Futuyu or Liu Hanxing cases (already published — cited above as prior findings). Source: domain-expert adjudication of the Shard 1–3 accepted pairs, scheduled Jul 16–21.]**

Finding 2 (a caption discrepancy or misattribution). **[TODO: Insert a NEW discrepancy between the archive's**

own metadata and the printed caption (place, date, subject, or photographer credit), with the same provenance-line and evidence requirements as Finding 1. State the discrepancy only; assert no editorial intent — intent is the historian's argument, made elsewhere.]

Finding 3 (optional; an editorial crop or retouch). [TODO: Insert only if adjudication yields it: a crop or retouch that changes the legible content of the printed image (e.g., a person or object removed by the crop), same evidence requirements. This slot is cut first if space is short.]

These findings cost the historian hours, not months: each begins from a review queue of a few hundred system-verified pairs rather than an archive of tens of thousands of photographs (see the workflow-impact analysis in the Human Evaluation subsection **[TODO: insert the measured adjudication throughput once logged during the Shard-3 campaign]**).

Ethics, Positionality, and Broader Impact

Two propaganda systems, one instrument. Both corpora are propaganda. The North China Railway Archive photographs were produced by the press apparatus of a Japanese corporate–military enterprise under occupation (1939–1945); the Jinchaji material was produced by the Chinese Communist Party's base-area pictorial unit, founded by the photographer Sha Fei. We treat the two systems even-handedly, as parallel objects of forensic description: the identical pipeline with identical frozen thresholds is applied to both, and the paper makes no claim about the relative truthfulness of either system. What it documents is a shared mechanism — the movement of photographs into recontextualized print — that both apparatuses operated.

Depicted persons and name restoration. The photographs show identifiable people living under occupation and at war. Where circulation evidence shows an attribution was erased in reprinting — a photographer's credit dropped, a named subject reduced to "common people" (Du, Le, and Honig 2024) — we follow a name-restoration principle: reattaching the documented name to the image is archival justice and is stated as fact; any claim about *why* the name was removed is interpretation and is deferred to historians.

The over-assertion risk. The system's core output — "these two images are the same exposure" — is exactly the claim that, made falsely, launders a misattribution into the scholarly record with a machine's authority. Three design consequences follow. Decision thresholds are set conservatively; every accepted pair is human-adjudicated before any historical claim is made (verification precision, conditional on the expert-labeled candidate set, is 90.2% on the held-out validation shard — roughly one accepted pair in ten is wrong before adjudication, which is why adjudication is not optional); and coverage is reported as a per-stage survival table (Table 4, **[TODO: numbers from the end-to-end audit over the 2,636 verified pairs, job W5]**) rather than implied to be complete.

Dual use. Nothing in the pose signal is specific to archives: a method that separates copies from re-shots of the same scene transfers to contemporary copy detection, misinformation triage, and copyright enforcement, and, like all instance-matching technology, could serve surveillance-adjacent uses. We release only archival material and annotations, and we state the technique's generality rather than leaving it implicit.

Dataset duties and licensing. The release ships with a datasheet (Gebru et al. 2021) that records collection provenance, the two-round verification protocol, and — centrally — the archive's own selection bias: the North China Railway Archive is itself a curated product of the occupation press apparatus, and what it omits is a property of the source that the benchmark inherits; the benchmark measures matching, not representativeness. North China Railway Archive images and metadata are published under CC BY 4.0 and are redistributed with attribution to the archive (Center for Open Data in the Humanities (CODH), ROIS 2019); for the Jinchaji side we release annotations and image identifiers only, no images.

Design policy. The Introduction's design principle — retrieval without interpretation — is implemented as policy, not aspiration: no output field of the system asserts editorial intent, and no intent language appears in the released annotations. Interpretation is reserved for the historian, whose findings appear — so labeled — in the preceding section.

Limitations

Image-only blindness. The pipeline decides on pixels, and the historically decisive manipulation is often textual: a recaption-only reprint is correctly matched as the same exposure, but the discrepancy that matters is invisible to the matcher, so caption changes are still found only by reading every adjudicated pair. **[TODO: Per-trace-type pipeline pass rates from the taxonomy tagging of confirmed positives (halftone-only, crop, rotation/mirror, retouch, recaption) — quantifies exactly which manipulation classes the image-only system cannot see.]** **Conditional metrics.** Verification P/R/F1 are conditional on the candidate set; the survival table is the paper's only end-to-end claim, and **[TODO: end-to-end survival numbers, audit W5]** bound what a historian ultimately misses. **Transductive training.** The fusion head is trained self-supervised on the full unlabeled working corpus, which includes the evaluation images (no labels are used). **[TODO: Control: retrain excluding Shard-3 images and show Shard-3 F1 unchanged within CI (report section 6.8, run W9); otherwise disclose as an uncontrolled limitation.]** **Annotation reliability.** The shard labels originate from a single annotation pass; **[TODO: Cohen's kappa from the independent dual-annotation under the written codebook, with disagreements adjudicated into gold labels and Unsure as a first-class category.]** **Degenerate geometry by design.** An aligned exact copy is a picture of a picture — near-planar, near-zero baseline — outside the training distribution of a 3D reconstruction model, and the near-zero inferred pose shift on true copies is a measured behavior, **[TODO: synthetic battery: pose-score distributions under controlled crops, rotations, halftone**

simulation, and mild perspective warps on ∼200 exact-copy pairs (report section 6.5, job W11)], not a guaranteed property; it is validated for one pinned checkpoint only. **Upstream crop dependency.** Every stage consumes YOLOv11 layout crops (Khanam and Hussain 2024); detection and crop-boundary errors propagate silently. **[TODO: Crop-stage QC: detection precision/recall on a 100-page audited sample, plus one sentence on failure handling.]**

Conclusion

We posed provenance verification for archival photographs as a question the copy-detection literature names but does not decide with geometry: same photograph, or same scene? Our contributions mirror that framing. We formulated exposure-identity vs. same-scene discrimination on cross-domain photo-to-halftone material and, to our knowledge, are the first to decide it using the relative camera pose inferred by a 3D foundation model, wrapped in a conservative, auditable decision layer. We release NCR-Match — 2,636 verified photo–print pairs with exposure-identity and same-scene-negative labels, with datasheet and code — deposited, not promised. With thresholds frozen on the development shard, the verifier reaches P 90.2 / R 98.4 / F1 94.1 on held-out validation, conditional on the candidate set (**[TODO: Shard-3 test row, one-shot frozen evaluation of Jul 22]**), and **[TODO: the Hokushi→Jinchaji cross-system transfer result, frozen thresholds, report section 6.7]** carries the same protocol across opposing propaganda systems. The findings of the preceding section — surfaced by the pipeline, adjudicated by a historian — are the intended unit of return. The pipeline retrieves; the argument belongs to the historian.

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